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Q1. CASE STUDY 1**CODE**

```
#include<stdio.h>
#include<string.h>

struct Project{
    char name[30];
    char role[20];
    int duration;
};

union status{
    char type[20];
    int months;
};

struct Emp{
    int id;
    char name[30];
    char dept[20];
    char desig[20];
    float salary;
    struct Project project;
    union status status;
    int isContract;
};

extern struct Emp globalDB[100];
struct Emp globalDB[100];

static void print(struct Emp e){
    printf("ID: %d | name:%s | dept: %s\n", e.id, e.name, e.dept);
    printf("Project : %s (%s) - %d mos\n", e.project.name, e.project.role, e.project.duration);
    if (e.isContract)
        printf("Status: Contract (%d months)\n", e.status.months);
    else
        printf("Status: full time (%s)\n", e.status.type);
}

int main(){
    globalDB[0].id=101;
    strcpy(globalDB[0].name, "Alice");
    strcpy(globalDB[0].dept, "IT");
    strcpy(globalDB[0].desig, "Developer");
    globalDB[0].salary = 75000.0;
    strcpy(globalDB[0].project.name, "CloudHR");
    strcpy(globalDB[0].project.role, "Lead");
    globalDB[0].project.duration =12;

    globalDB[0].isContract=0;
    strcpy(globalDB[0].status.type, "Permanent");
    print(globalDB[0]);
    return 0;
}
```

OUTPUT

(no output)

Q2. CASE STUDY 2

CODE

```
#include<stdio.h>
#include<string.h>

struct Tester{
    int id;
    char name[30];
};

union status{
    char activenote[30];
    int progresspercent;
    char closeddate[20];
};

struct bug{
    int bugID;
    char desc[50];
    char severity[10];
    char developer[30];

    struct Tester tester;
    union status status;
    int statusCode;
};

int main(){
    struct bug b1;
    b1.bugID= 101;
    strcpy(b1.desc, "Login page button fails on click");
    strcpy(b1.severity, "High");
    strcpy(b1.developer, "Alix");

    b1.statusCode=2;
    b1.status.progresspercent=60;
    printf(" bug report\n");
    printf("bug id: %d | Severity: %s\n", b1.bugID, b1.severity);
    printf("Description:%s\n", b1.desc);
    printf("Assigned Developer: %s\n", b1.developer);
    printf("Reported by tester: %s (id:%d) \n", b1.tester.name, b1.tester.id);
    if(b1.statusCode==1)
        printf("Status:active (%s)\n", b1.status.activenote);
    else if(b1.statusCode ==2)
        printf("Status: In Progress (%d%% completed)\n",b1.status.progresspercent);
    else if(b1.statusCode==3)
        printf("Status:Closed on %s\n", b1.status.closeddate);
}
```

OUTPUT

(no output)

Q3. CASE STUDY 3

CODE

```
#include<stdio.h>
#include<string.h>

struct supplier{
    int id;
    char name[30];
};

union categorydetails{
    int warrentyMonths;
    char size[10];
    char expirydate[15];
};

struct product{
    int id;
    char name[30];
    float price;
    int stock;
    struct supplier supplier;
    union categorydetails cat;
    int type;
};

int main(){
    struct product p1;
    p1.id=101;
    strcpy(p1.name, "laptop");
    p1.price =75000.50;
    p1.stock=15;
    p1.supplier.id=501;
    strcpy(p1.supplier.name, "TechCorp");
    p1.type = 1;
    p1.cat.warrentyMonths=24;
    printf("  Prodcut cateflog\n\n");
    printf("product id:%d |Name: %s\n", p1.id, p1.name);
    printf("Price:Rs.%.2f | Stock: %dpcs \n", p1.price, p1.stock);
    printf("Supplier:%s (ID: %d)\n", p1.supplier.name, p1.supplier.id);

    if(p1.type ==1)
        printf("Category :electronics(warrenty:%d months)\n", p1.cat.warrentyMonths);
    else if (p1.type ==2)printf("category:Clothing size:%s \n", p1.cat.size);
    else if( p1.type== 3)printf("Category: grocery (Expiry: %s)\n", p1.cat.expirydate);
    return 0;
}
```

OUTPUT

(no output)

Q4. CASE STUDY 4

CODE

```
#include<stdio.h>
#include<string.h>
struct visit{
    char diag[50];
    char doc[30];
    float cost;
};
union statusdetails{
    int roomnumber;
    char dischargedate[15];
    int observationhours;
};
struct patient{
    int id;
    char name[50];
    int age;
    char bloodgroup[10];
    char history[50];
    struct visit visit;
    union statusdetails status;
    int statustype;
};
int main(){
    struct patient p1;
    p1.id=101;
    strcpy(p1.name, "John Doe");
    p1.age=45;
    strcpy(p1.bloodgroup, "+O");
    strcpy(p1.history, "Hypertension0");

    strcpy(p1.visit.diag, "Acute Fever");
    strcpy(p1.visit.doc, "Dr.Smith");
    p1.visit.cost=150.00;
    p1.statustype=1;
    p1.status.roomnumber=302;
    printf("    Patient Record  \n\n");
    printf("ID:%d | Name:%s |age: %d | blood Group: %s\n", p1.id,p1.name, p1.age, p1.bloodgroup);
    printf("Medical History: %s\n", p1.history);
    printf("Doctor: %s | Diagnosis: %s | Cost: Rs.%.2f\n", p1.visit.doc, p1.visit.diag, p1.visit.cost);

    if(p1.statustype==1)
        printf("Status: admitted (room No: %d)\n", p1.status.roomnumber);
    else if(p1.statustype==2)
        printf("Status: Disccharge(Date:%s)\n", p1.status.dischargedate);
    else if(p1.statustype==3)
        printf("Status: Under Observation (%d hours reminnaing)\n", p1.status.observationhours);
    return 0;
}
```

OUTPUT

(no output)